John (Ioannis) Maris,

Data Science M.Sc.

I'm a data scientist with a strong foundation in mathematics, statistics, and machine learning. My experience covers both industry and academia, working at TOYOTA MOTOR EUROPE in Brussels, Belgium, applying data science in the automotive sector, and advancing state-of-the-art generative AI for protein engineering at FORTH-Hellas, with a scholarship funded by Apple.

I hold a B.Sc. in Mathematics & Applied Mathematics and an M.Sc. in Data Science & Machine-Statistical Learning, graduating as Valedictorian. I specialize in deep learning, transformer-based generative models, and time series forecasting. I focus on building robust, practical solutions that connect theory with real-world impact.

About Me

Brussels, Belgium

+30 6987328453

gianismaris13@gmail.com

Portfolio

LinkedIn

GitHub

Professional Experience

Data Scientist, BEVs | R&D

TOYOTA MOTOR EUROPE - I*ntern - Connected Powertrain* | *Dec 2024 - June 2025*

- Built a range recommendation system for BEVs by predicting energy consumption using real-time big data, trained on hundreds of real trips across Europe for the Toyota bZ4X.
- Used data-driven modeling with control signals and physical variables to forecast vehicle range under varying conditions.
- Successfully presented the findings to senior leadership at Toyota Motor Corporation in Japan, contributing to product development insights.

Generative AI Engineer | R&D

FORTH-Hellas (IACM Lab) | Mar. 2024 - July 2025

- *Developed deep generative AI models (e.g., VAEs, GANs, and diffusion models) to generate new protein sequences targeting plastic degradation and antimicrobial resistance.*
- *Used DeepMind's AlphaFold3 to validate structural viability and Meta's ESM to extract embeddings.*
- *Contributed to a scalable pipeline for AI-driven protein engineering.*

Statistical Analyst | R&D

FORTH-Hellas - Intern | Dec. 2022 - July 2023

- *Built predictive models for biomedical datasets using regularized statistical methods*
- *Developed automated tools for model selection and tuning*
- *Improved reproducibility and interpretability in high-dimensional analysis across multiple research projects*

University Teaching Assistant

University of Crete | Lectures, Tutoring, Assigning and Grading Exams. | Sep. 2022 - June 2024

Tutored large classes of Master's and undergraduate students.

- *Machine Learning (Postgraduate) | Python Computer Language (Fall 2023)*
- *Introduction to Linear Algebra (Fall 2022)*
- *Numerical Analysis (Spring 2024)*

Educational Background

Master of Sciences in Data Science & Machine-Statistical Learning
University of Crete & FORTH, Greece | 110 ECTS | Sep. 2023 - July. 2025 | GPA 9.0 | Valedictorian.
- Focus on Machine Learning, Deep Generative Models, and Statistical Inference.
- Courses in Time Series Forecasting, Statistics, NLP & LLMs, Bayesian Inference, Big Data, Data Structures.
- Thesis on '*Generative AI for Protein Engineering using Large Language Diffusion Models'.*

Bachelor of Sciences in Mathematics & Applied Mathematics
University of Crete, Greece | 274 ECTS | Sep. 2017 - Nov. 2022 | GPA 7.6
- Specialized in Statistics, Programming, and Applied Mathematics.
- Courses in Linear Algebra, Optimization, Probability & Statistics, Numerical Analysis, Programming, Fourier Analysis.
- Thesis on '*Supervised Classification with Parametric Models'.*

Honors & Awards

Apple MSc. Scholarship.
'Disentangled Representation Learning via Mutual Information Optimization.' (Nov. 2024).
- Awarded from Apple, through IACM FORTH for my research in protein engineering to develop enzymes for plastic degradation

Erasmus+ Traineeship Grant. (3 months)
- Awarded from European Commission though Technical University of Crete.

Erasmus+ Traineeship Grant. (4 months)
- *Awarded from European Commission though University of Crete.*

IPTO Scholarship.
- *Awarded by Independent Power Transmission Operator for my MSc research on genAI protein engineering for plastic degradation.*

Projects & Publications

In Proceedings of 24th international congress on acoustics, ICA, Acoustical Society, Korea.
- *'Identification of Normal Modes in Underwater Acoustic Propagation using Convolutional Neural Networks'.*

Authors: Costas Smaragdakis, John Maris, Michael Taroudakis. (2022)

M.Sc. Thesis. (2025)
- *'Generative AI in Protein Engineering using Large Language Diffusion Models'.*

B.Sc. Thesis (2023)
- *'Supervised Classification with Parametric Models for Identifying Tumorous Breast Cancer Patients'.*

FORTH-HELLAS (Time Series Analysis) (2024)
- *'15-Minute Ahead Traffic Forecasting in Athens Using Robust Time Series Models'.*

Community & Interests

- Fluent in English & Greek.
- Elementary Japanese.
- Mathematical competitions
- Chess high elo player.
- Politically oriented.
- Enthusiast in stock market analysis, Finance, and Astronomy.
- Interested in philosophy and history.
- Cooking and experimenting with international cuisine.

References

- Dr. Yiannis Kamarianakis

Principal Researcher at IACM-FORTH.
Former Professor at Arizona State University.
Supervisor during my internship.

- Julie Le Louvetel

Manager at TOYOTA MOTOR EUROPE.

- Julien Bouilly

Technical Manager at TOYOTA MOTOR EUROPE.

- Tijl Janssens

Engineer at TOYOTA MOTOR EUROPE, and
Supervisor during my internship.

- Dr. Yiannis Pantazis

Principal Researcher at IACM-FORTH.
Former Postdoctoral Research Associate at the University of Massachusetts.

- Dr. Michael I. Taroudakis

Professor Emeritus at University of Crete.